Performance Evaluation of Bankruptcy Prediction Models: An Orientation-Free DEA-based Framework

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Abstract: Prediction of corporate failures is one of the major activities in auditing firms' risks and uncertainties. The design of reliable models to predict bankruptcy is crucial for many decision making processes. Although a large number of models have been designed to predict bankruptcy, the relative performance evaluation of competing prediction models remains an exercise that is unidimensional in nature. To be more specific, although most studies use several performance criteria and their measures to assess the relative performance of bankruptcy prediction models, the assessment exercise of competing prediction models is generally restricted to their ranking by measure, which leads to reporting conflicting results. In this research, we overcome this methodological issue by proposing an orientation-free data envelopment analysis (DEA) model as a multi-criteria assessment framework and report numerical results based on UK data.

Keywords: Bankruptcy Prediction; Performance criteria; performance measures; Data Envelopment Analysis; Slacks-Based Measure

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1. Introduction

Corporate failure often occurs when a firm experiences serious losses and/or becomes insolvent with liabilities that are disproportionate to its assets. Corporate failure may result from one or a combination of internal and external factors; e.g., managerial errors due to insufficient or inappropriate industry experience, risk seeking managers, lack of commitment and motivation to lead the company efficiently, refusal or failure to adjust managerial and operational structures of the firm, inefficient or inappropriate corporate policies, economic climate, changes in legislation, industry decline – see for example Van Gestel et al. (2006). Bankruptcy induces substantial costs to the business community such as court costs, lawyer costs, lost sales, lost profits, higher cost of credit, inability to issue new securities and lost investment opportunities (e.g., Altman, 1968, Premachandra et al., 2009, Van Gestel et al., 2006). Therefore, the design of reliable models to predict bankruptcy is crucial to audit business risks and assist managers to prevent the occurrence of failure, and assist stakeholders to assess and select firms to collaborate with or invest in (e.g., Ahn et al., 2000, Balcaen and Ooghe, 2006). Given the importance of bankruptcy prediction, there is a considerable amount of literature focusing on both financial and non-financial information, and proposing new bankruptcy prediction models to classify firms as healthy or non-healthy (e.g., Dimitras et al., 1999, Balcaen and Ooghe, 2006, Ravi Kumar and Ravi, 2007). With the increasing number of quantitative models available, one of the challenging issues faced by both academics and professionals is how to evaluate these competing models and select the best one(s) (Balcaen and Ooghe, 2006). Our survey of the literature on bankruptcy prediction revealed that although most studies tend to use several performance criteria (e.g., discriminatory power, calibration accuracy, and correctness of categorical prediction) and, for each criterion, one or several measures (e.g., Kullback-Leibler (KL) distance, Kolmogorov Smirnov (KS), area under the receiver operating characteristic curve (AUC), Brier score and/or Binomial, Type I and Type II errors) to evaluate the performance of competing prediction models, the assessment exercise is generally restricted to the ranking of models by measure. For example, Hillegeist et al. (2004) compared the performance of multivariate discriminant analysis (MDA) and Black-Scholes-Merton (BSM) – based models using log likelihood, Pseudo-$R^{2,}$ Vuong tests and information content tests. The overall results were inconsistent because of the unidimensional nature of their rankings. Bandyopadhyay (2006) used the MDA and logit regression models to predict corporate failures in Indian. They used both discriminatory power

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and correctness of categorical prediction criteria and several metrics (i.e., receiver operating characteristic curve (ROC), Type I and Type II errors, and overall correct classification (OCC)), but their results were mixed with respect to which model outperformed the rest as a result of the unidimensional nature of their rankings. Agarwal and Taffler (2008) predicted the UK bankrupted firms using MDA and market-based models. They compared the relative performance of these two classes of models based on a range of measures (e.g., AUC, accuracy ratios, economic value of different models, Clark test and Vuong tests), but they did not identify a constantly superior model with regards to different measures.

There are several issues with performance evaluation of bankruptcy prediction models. For example, many studies restricted their performance evaluation exercise to the use of a single criterion and one or several measures of that criterion and ranked models by measure. On the other hand, other studies considered more than one criterion and for each criterion one or several measures but again the performance of models is restricted to their ranking by measure. In sum, the performance evaluation exercise whether it considers a single criterion or multiple criteria and whether a single measure or multiple measure of a criterion are used, the performance evaluation remains unidimensional in nature, on one hand, and the big picture is not taken into account in that a single or a very restricted number of criteria only are used, on the other hand – see Appendix A for a summary of the literature on performance criteria and measures used in evaluating bankruptcy prediction models.

The drawback of the commonly used approach for the relative performance evaluation of competing bankruptcy prediction models is that the rankings corresponding to different criteria are often different, which result in a situation where one cannot make an informed decision as to which model performs best when taken all criteria and their measures into consideration. In this paper, we intend to fill this gap by proposing a data envelopment analysis (DEA)-based framework for the relative performance of bankruptcy prediction models. The key contribution of this paper is to propose a systematic evaluation model to help the forecasting community with ranking a set of competing prediction models under multiple criteria. Furthermore, we provide a general classification of performance criteria for prediction models of categorical variables along with an exhaustive list of performance measures. Finally, we test the performance of a popular category of bankruptcy prediction models; namely, statistical models, using the most popular criteria along with a relatively large number of measures of these criteria.

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The remainder of this paper is organized as follows. In section 2, we survey and classify the literature on bankruptcy prediction models. In section 3, we survey and classify the literature on the performance criteria of bankruptcy prediction models and their metrics. In section 4, we present the proposed multi-criteria methodology; namely, an orientation-free super efficiency DEA framework to evaluate the relative performance competing forecasting models for predicting bankruptcy. In section 5, we present and discuss our empirical findings. Finally, section 6 concludes the paper.

2. Bankruptcy Prediction Models

Bankruptcy prediction models could be divided into two main categories; namely, statistical models and non-statistical models. Statistical models could be further divided into four sub-categories; namely, discriminant analysis models, regression models for categorical variables, survival analysis models and market-based models. On the other hand, non-statistical models could be further divided into five sub-categories; namely, nearest neighbour models (e.g., Tam and Kiang, 1992), classification trees-based models (e.g., Breiman et al., 1984, Arminger et al., 1997), mathematical programming-based models (Freed and Glover, 1981), and artificial intelligence models (e.g., Boser et al., 1992, Bryant, 1997). In this paper, we focus on the relative performance of the most widely used statistical models that have been used in the literature on bankruptcy prediction (see for example, Hillegeist et al. 2004; Bandyopadhyay, 2006; Agarwal and Taffler (2008). Hereafter, we provide a brief description of these models and discuss the main similarities and differences between sub-categories of statistical models.

2.1 Discriminant Analysis Models

Discriminant Analysis (DA) – first proposed by Fisher (1938), is a collection of classification methods which aim at partitioning observations into two or more subsets or groups so as to maximize within-group similarity and minimize between-group similarity, where "similarity" is measured by some sort of distance between observations (e.g., Mahalanobis distance). Univariate DA was first applied to bankruptcy prediction by Beaver (1966) and MDA was first applied to bankruptcy prediction by Altman (1968). A generic MDA model could be summarized as follows:

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$$z = f\left(\sum_{j=1}^{P} \beta_j x_j\right),$$

where z is commonly referred to as a score, x_js are explanatory variables, β_js represent the coefficients of the explanatory variables in the model, and f denotes the mapping of $\beta^t \cdot x$ on the set of real numbers $\Re$ - often referred to as a classifier, and could be either linear or non-linear. Note that in comparing MDA models to other sub-categories of statistical models, one would typically need to estimate the probability of default (PD), which is used as an input to very many performance measures. In this paper, we follow Hillegeist et al., (2004) in using a logit transformation:

$$PD = \frac{e^{z-\text{score}}}{1 + e^{z-\text{score}}}$$

Note that, under the normality assumption, MDA and logit approaches are closely related Hekanaho et al. (1998). For a two-group classification problem, the classifier f is often a simple function that maps all observations or cases with discriminant or z-score values above a certain threshold or cut-off point to the first group and all other cases to the second group, where the cut-off point – often referred to as the *cutting score* or the *critical* z*-score*, is the average of groups' centroids, if group sizes are equal, or their weighted average, if group sizes are unequal, where a *group centroid* refers to the vector of group means of the explanatory variables. Note that when the cost of misclassifying an object into the wrong group differs from one group to another, the cutting score should be designed so as to minimize the cost of misclassification. From an application perspective, the main limitation of this methodology lies in the nature of the underlying assumptions; namely, in each group of observations, explanatory variables are normally distributed; the variance-covariance matrices are equal across classification groups – also referred to as the homoscedasticity assumption; and the prior probabilities of belonging to each group as well as misclassification costs are known or pre-specified. In fact, empirical findings often suggest that the normality assumption as well as the homoscedasticity assumption are violated (Balcaen and Ooghe, 2006), which affect the prediction ability of an MDA model. As to the last assumption, in practice, both prior probabilities of belonging to each group and misclassification costs are unknown and their pre-specification would both introduce bias along with a degree of arbitrariness into the choice of the cut-off point. Furthermore, in practice, multicollinearity among explanatory variables (e.g., financial ratios) is often an issue, as it affects

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the estimates of the weights and makes their interpretation unreliable. In the literature on bankruptcy prediction, MDA models mainly differ with respect to the choice of the explanatory variables and the form of the classifier – see Appendix A, and are part of most comparative analysis exercises.

2.2 Regression Models for Categorical Variables

As compared to discriminant analysis, regression models for categorical variables – also known as probability models (e.g., logit, probit, gompit) allow one to overcome some of the limitations of discriminant analysis. For example, within a regression framework for discrete response variables, the normality and the homoscedasticity assumptions are relaxed, on one hand, and the knowledge of prior probabilities of belonging to each group as well as misclassification costs is not required, on the other hand. The generic model for binary variables could be stated as follows:

$$\begin{cases} PD = \Pr ob(y=1) = 1 - \Pr ob(y=0) \\ PD = F(\beta, x) \end{cases}$$

where y denotes the categorical response variable, x denotes the vector of explanatory variables, β denotes the vector of coefficients of x in the model, and F is a function – commonly referred to as the link function, that maps any real number; e.g., score $\beta^t x$, onto a probability. The choice of F determines the type of probability model. For example, the normal probability model – known as probit, assumes that the link function is the cumulative normal distribution, say Φ; that is, $F(\beta, x) = \Phi^{-1}(\beta^t x)$ or equivalently:

$$PD = \Phi(\beta^t x) = \int_{-\infty}^{\beta^t x} \phi(t)dt,$$

where ϕ denotes the standard normal density function. The logistic probability model – known as logit, assumes that the link function is the cumulative logistic distribution function, say Λ; that is, $F(\beta, x) = \Lambda^{-1}(\beta^t x)$ or equivalently:

$$PD = \Lambda(\beta^t x) = \frac{e^{\beta^t x}}{1 + e^{\beta^t x}}.$$

The extreme value probability model – known as gompit, assumes that the link function is the cumulative Type I extreme value distribution function, say E; that is, $F(\beta, x) = E^{-1}(\beta^t x)$ or equivalently:

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$$PD = E(\beta^t x) = \exp\left(-e^{-\beta^t x}\right).$$

First proposed by Cox (1972), logistic regression has first been applied to bankruptcy of banks by Martin (1977) and to bankruptcy of non-financial firms by Ohlson (1980). The assumptions underlying logit models could be summarized as follows. First, the response variable need not be normally distributed, but its distribution is within the range of the exponential family of distributions; e.g., Normal, Poisson, Binomial, or Gamma. Second, the explanatory variables are linearly related to the logit of the probability of the event or group of interest. Finally, residuals are statistically independent and binomially distributed with mean zero and variance $p(x)(1-p(x))$. Note that, when the second assumption is violated, logistic regression will underestimate the degree of relationship of the independents to the dependent and will lack power; thus, generating Type II errors, thinking there is no relationship when there actually is one. As alternative models, one could use probit or gompit. In the literature on bankruptcy prediction, logit is the most popular probability model and logit models only differ with respect to the choice of the explanatory variables – see Appendix A, and are part of most comparative analysis exercises.

2.3 Survival Analysis Models

Discriminant analysis models as well as probability models (e.g., logit, probit, gompit) are cross-sectional models and as such fail to take account of differences in firms' performance or risk profile over time; in sum, the probability of default (PD) provided by these static models is time-independent. In order to overcome this issue, one could use a dynamic methodology such as survival analysis. Survival Analysis is concerned with the analysis of time to events. In this paper, we limit ourselves to a single event of interest; namely, bankruptcy or failure. Two functions are of special interest in survival analysis; namely, the survival function and the hazard function. The survival function, say $S(t)$, is a function of time and represents the probability that the time of failure is later than some specified time t; that is: $S(t) = P(T \succ t)$, where T is a random variable describing the time of failure for an observation or firm; in sum, the survival function provides survival probabilities or the probabilities of survival past specified times. On the other hand, the hazard function, say $h(t)$, is also a function of time and represents the failure or hazard rate at time t conditional on survival until t or later; that is:

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$$h(t) = \lim_{\Delta t \to 0} \frac{P(t \le T \le t + \Delta t | T \ge t)}{\Delta t} = -\frac{S'(t)}{S(t)},$$

where $S'(t)$ denotes the derivative of the survival function S with respect to time. As far as the bankruptcy prediction application of survival analysis is concerned, the aim is to model the relationship between survival time and a set of explanatory variables. The most commonly used model is the proportional hazards model – also known as the Cox regression model, proposed by Hekanaho et al. (1998) and used to model bankruptcy for the first time by Shumway (2001). The proportional hazards model with time-dependent explanatory variables could be stated as follows:

$$h(t, x_t) = h_0(t) e^{\beta^t x_t},$$

where x_t denotes the vector of explanatory variables in period t, β denotes the vector of coefficients of x in the model, $h_0(t)$ denotes the unconditional hazard function – commonly referred to as the baseline hazard, and the response variable $h(t, x_t)$ represents the failure or hazard rate conditional on x_t. The survival function corresponding to the proportional hazards model with time-dependent explanatory variables is:

$$S(t, x_t) = e^{-\int_0^t h(t, u_t) du_t}.$$

According to Shumway (2001) the likelihood function of both hazard models and logit models are the same which results in the same asymptotic variance-covariance matrix. So we replaced the hazard model with a logit model in which an information matrix including all historical values of the explanatory variables for a firm is used instead of each firm-year as an independent observation. Therefore, the probability of default at time period t is:

$$PD = \frac{e^{h_0(t) + \beta^t x}}{1 + e^{h_0(t) + \beta^t x}}$$

2.4. Market-based models

Discriminant analysis models, probability models and survival analysis models are also known as the accounting-ratio-based models, as these models typically built by searching through a large number of accounting ratios which the ratio weightings estimated on a sample of healthy and non-healthy firms. Despite their simplicity and popularity, these models are suffering from a number of issues. First, the accounting statements present a firm's historical performance

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and may not be informative in predicting the future. Second, the "true" asset values may be very different from the book values. In addition, accounting numbers can be manipulated by the management (e.g., Balcaen and Ooghe, 2006; Agarwal and Taffler, 2008). In order to overcome these drawbacks, one could make use of market-based models, which explicitly make use of market variables. The rational of this class of model is that in an efficient market, stock prices will reflect both the information contained in the accounting statements as well as the future expected cash-flows. Furthermore, market variables are unlikely to be influenced by firm's accounting policies. However, the potential drawbacks for market-based models are the requirement of the normality assumption of stock returns, and it does not distinguish between different types of debt and assumes that the firm only has a single zero coupon loan.

The key idea of a market-based model is that holding the equity of a firm is as like take a long position in a call option. The probability of bankruptcy is the probability that the call option will expire worthless, or the value of the assets is less than the face value of the liabilities at the end of the holding period. In this paper, we focus on two popular types of market-based models and both are based on the BSM contingent claims model proposed by Hillegeist et al. (2004) and Bharath and Shumway (2008) respectively. Both models follow McDonald et al. (2006) approach in estimating the likelihood that the value of assets at the end of holding period would be less than the face value of debt:

$$\mathrm{PD}=\mathrm{N}\left(-\frac{\ln\left(\mathrm{V_A}/\mathrm{X}\right)+\left(\mu\text{-}\delta\text{-}0.5\times\sigma_{\mathrm{A}}^{2}\right)\times\mathrm{T}}{\sigma_{\mathrm{A}}\sqrt{\mathrm{T}}}\right)$$

where V_A and σ_A denote the asset value and volatility respectively, μ and δ are referred to as expected return of the firm and dividend rate respectively and $N(\ )$ represents the cumulative normal density function as a link to estimate a probability between 0 and 1. Note that V_A , σ_A and μ are not direct observable and the key different between Hillegeist et al. (2004) and Bharath and Shumway (2008)'s approach is how they estimate these values – the reader is referred to Appendix A for details on these models.

3. Performance Criteria and Measures

In this section, we classify and summarize the literature on performance criteria of bankruptcy prediction models and their measures. In this paper, we classify performance criteria

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into six categories; namely, reliability, costs, benefits, complexity, universality, and ability to incorporate managerial judgement. We define *reliability* as a multidimensional criterion consisting of four sub-criteria; namely, theoretical relevance, validity, accuracy, and stability. The *theoretical relevance* sub-criterion refers to the degree of suitability of a model for a given data set and a given forecasting exercise; that is, its ability, from a design perspective, to take into account all features of the data set under consideration as well as its suitability for a specific forecasting horizon – the relevance of the forecasting horizon lies in the fact that, by design, some forecasting models are more suitable for a specific time horizon than others. The *validity* sub-criterion refers to whether the assumptions underlying a model hold or not. From a methodological perspective invalid models should be discarded from further consideration; however, in practice things are not either black or white. For example, within a discriminant analysis framework, one needs to test whether in each group of observations, explanatory variables are normally distributed or not – using several normality tests which may lead to different conclusions; in this case, one may conclude that the normal distribution is a reasonable approximation and consider the model as valid. The *accuracy* sub-criterion refers to the ability of a model to reproduce the past and to forecast the future – it is a multidimensional construct that has three main dimensions; namely, the *discriminatory power* dimension, the *calibration accuracy* dimension and *correctness of categorical predictions* dimension. The *discriminatory power* dimension refers to the ability of model to discriminate between the good cases and the bad cases, where a case is referred to a firm. The *calibration accuracy* dimension refers to the quality of estimations of the probability of default (PD) and the loss given default (LGD). The *correctness of categorical prediction* dimension refers to the ability of a model to produce forecasts that are consistent with actual in that forecasts reveal firms as healthy (respectively, non-healthy) when actual is healthy (respectively, non-healthy) – this criterion is particularly important for decision makers. The *stability* sub-criterion refers to the extent to which the estimates of model coefficients are stable in that they are statistically significant (i.e. determined with high confidence – 1% or 5%, and low uncertainty), make a significant contribution to the model (i.e., inclusion of a variable in the model yields a significant improvement in its deviance), and are not too dependent on the data sample (e.g., results on performance characteristics such as area under the curve (AUC) are similar both in- and out-of-sample). The second criterion in our classification is *cost* refers to the relevant costs; e.g., the cost of developing or purchasing and

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maintaining a forecasting system, the cost of purchasing or collecting data and its storage and pre-processing, the costs related to the time required to obtain a forecast, the costs of repeated applications or use of a model to produce forecasts, etc. The third criterion is *benefit* and refers to the improvements resulting from improved decisions such as direct and indirect cost savings. The fourth criterion is *complexity* and refers to the extent or degree to which the model or method is easy to understand by users/managers; extent to which it is easy to interpret results; e.g., model coefficients – also referred to as the readability sub-criterion; level of conceptual and technical knowledge or expertise required for an effective use of the model, etc. The fifth criterion is *universality* and refers to the extent to which the model is widely used in practice or relevant users are familiar with; the extent to which the model is available within popular software packages, etc. The sixth and last criterion is the *ability to incorporate managerial judgement* and refers to the extent to which the model or method provides for means to incorporate subjective judgement.

In the literature on credit scoring and bankruptcy prediction, the most commonly used criterion is accuracy; therefore, hereafter we shall restrict our attention to the measures of this criterion. The measures of the first sub-criterion of accuracy; namely, *discriminatory power*, could be divided into two categories: statistics based on the differences in expectations of the distribution functions of the scores of the good cases and the bad cases, and statistics based on the differences between the distribution functions of the scores of the good cases and the bad cases. The statistics based on the differences in expectations of the distribution functions of the scores of the good cases and the bad cases could be further divided into two sub-categories; namely, 1) *measures concerned with comparing expectations of distributions at a given point in time* (e.g., Kullback-Leibler (KL) distance also referred to as the divergence statistic, Conditional Information Entropy Ratio (CIER), Information value (IV), Mahalanobis Distance (D_M)), and 2) *measures concerned with comparing expectations of distributions at different points in time* (e.g., Stability Index). On the other hand, the statistics based on the differences between the distribution functions of the scores of the good cases and the bad ones could also be further divided into two sub-categories; namely, 1) *measures concerned with comparing cumulative distribution functions (CDFs) of scores of the good cases and the bad cases* (e.g., Kolmogorov Smirnov (KS) Statistic) and 2) *measures based on the features of graphs concerned with comparing CDFs of scores of the good cases and the bad cases* (e.g., Area under Receivable

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Operating Characteristic (AUROC) – also known as c-statistic, Gini Coefficient – also known as Gini Index or Somer's D, Pietra Index, Accuracy Ratio (AR)). The measures of the second sub-criterion of accuracy; namely, *calibration*, are typically measures related to statistical tests such as the Binomial test, the Chi-square test – often referred to as the Hosmer-Lemeshow (HL) test, the log-likelihood test, and the Spiegelhalter test and the corresponding Brier Score (BS). As to the third sub-criterion of accuracy; namely, *correctness of categorical predictions*, it could be measured by sensitivity, specificity, Type I and Type II errors, misclassification rate (MR), overall correct classification (OCC) etc. The reader is referred to Appendix B for details on these measures.

4. A Slacks-based DEA Framework for Assessing Bankruptcy Prediction Models

In this paper, we propose a DEA-based framework for assessing the relative performance of competing bankruptcy prediction models. In this section, we first present the basic concepts and models of DEA. Next, we discuss how one might adapt a DEA framework to measure and evaluate the relative performance of competing bankruptcy prediction models.

4.1. Basic concepts and models

DEA is a mathematical programming-based approach for assessing the relative performance of a set of decision making units (DMUs), where each DMU is viewed as a system and defined by its inputs, its processes, and its outputs. The basic optimization problem addressed by DEA may be stated as follows:

> *Basic DEA Optimization Problem: Maximize the performance of a given DMU – as measured by the ratio of a weighted linear combination of outputs to a weighted linear combination of inputs, under the constraints that such ratio is less than or equal to one for each DMU and the weights are non-negative.*

Using the Charnes-Cooper transformation (Charnes and Cooper, 1962, Charnes et al., 1978) , the mathematical programming formulation of this basic optimization problem is transformed into a linear program and therefore is easy to solve. The mathematical formulations of the basic DEA input- and output-oriented linear optimization problems are presented in Table 1, where $x_{i,j}$ denotes the amount of input i used by DMU_j, $y_{r,j}$ denotes the amount of output r produced by DMU_j, v_i (respectively u_r) denotes the weight of input i (respectively, output r), and these

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weights are the decision variables. Note that DEA models where the decision variables are the weights of input and output quantities are said to be stated in a multiplier form. Note also that the optimal value of e_k^{input} (respectively, e_k^{output}) indicates the efficiency status of DMU_k; to be more specific, $e_k^{input} = 1$ (respectively, $e_k^{output} = 1$) means that DMU_k is efficient in that its weighted sum of outputs is equal to its weighted sum of inputs, and $e_k^{input} \prec 1$ (respectively, $e_k^{output} \succ 1$) means that DMU_k is inefficient in that it produces less output than the input it requires. The set of efficient DMUs is referred to as the efficient frontier and represents the empirical standard of excellence.

[INSERT Table 1 HERE]

In general, the duals of these multiplier problems – commonly referred to as envelopment problems, are typically used in a relative performance evaluation exercise. The mathematical formulations of the basic DEA input- and output-oriented envelopment problems are presented in Table 2, where the variable θ_k is the dual variable associated with the fixed output amount constraint in the primal and may be interpreted as the technical efficiency ratio of DMU_k, the variable λ_j is the dual variable associated with the technical efficiency ratio of DMU_j constraint in the primal and may be interpreted as the weight assigned to DMU_j's inputs and outputs in constructing the ideal benchmark of DMU_k, the first set of constraints of, for example, the input-oriented envelopment model state that, for each input i, the amount used by DMU_k's "ideal" benchmark; that is, the projection of DMU_k on the efficient frontier, should at most be equal to the "revised" amount used by DMU_k; i.e., amount adjusted for the degree of technical efficiency θ_k of DMU_k, and the second set of constraints of, for example, the input-oriented envelopment model state that, for each output r, the amount produced by DMU_k's "ideal" benchmark; that is, the projection of DMU_k on the efficient frontier, should be at least as large as the amount produced by DMU_k. It is obvious that envelopment models allow for more appealing interpretations. In addition, it is not always easy to compute an excess in an input or a shortage in an output from the optimal solution of a model expressed in multiplier form, whether input- or output-oriented; however, solving the dual would enable one to determine excesses and shortfalls explicitly by the non-zero values of the slack and surplus variables.

[INSERT Table 2 HERE]

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Last, but not least, DEA models – whether expressed in multiplier form or in envelopment form allow one to identify the reference set or peer group used to benchmark each DMU in seeking improvements; for example, the reference set of a specific DMU, say DMU_k, is the set of DMUs with positive dual variables λ_j. Note that if the optimal value of θ_k, say θ_k^*, (respectively ϕ_k, say ϕ_k^*) is equal to 1, then the DMU_k under evaluation is efficient; else, $\theta_k^* \prec 1$ (respectively, $\phi_k^* \succ 1$) indicates that DMU_k is inefficient and the current level of inputs (respectively, outputs) should be decreased (respectively, increased). As our objective is to provide a multidimensional ranking to get rid of the inconsistencies of unidimensional rankings, we only make use of DEA efficiency scores. For a detailed discussion of different DEA models and its application areas, the reader is referred to Zhu (2003), Cooper et al. (2004, 2007), Cook and Seiford (2009), and Liu et al. (2013).

4.2. Adaption of DEA framework

DEA is a generic framework and as such its implementation for this specific relative performance evaluation exercise requires a number of key decisions to be made.

First, *what are the units to be assessed or DMUs?* In this paper, DMUs are fourteen competing statistical bankruptcy prediction models – see Appendix A for a general description of these models.

Second, w*hat are the inputs and the outputs?* The inputs and outputs are the relevant performance criteria, along with their measures, to be used for assessing bankruptcy prediction models. Here we focus on the discriminatory power, the calibration accuracy and correctness of categorical predictions. In this paper, discriminatory power is measured by one of the following measures: AUC, Gini coefficient, KS, CIER and IV; calibration accuracy is measured by BS and HL test; and correctness of categorical predictions is measured by Type I errors, Type II errors, MR and OCC. Within the proposed DEA framework, inputs are to be minimized and outputs are to be maximized; consequently, any measures of calibration accuracy and correctness of categorical predictions could be used as input, and any measure of discriminatory power could be used as output - the reader is referred to Appendix B for details on measures of criteria.

Third, *what is the appropriate DEA mathematical program to solve?* Although basic DEA models could be used to classify competing forecasting models into efficient and non-efficient

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ones and rank them according to their scores, one cannot differentiate between efficient ones as they all receive a score of 1. In many application areas, decision makers are interested in obtaining a complete ranking in order to refine DMUs evaluation and our application is no exception. In this paper, we propose an orientation-free super-efficiency DEA framework; namely, a slacks-based super-efficiency DEA framework for assessing the relative performance of competing bankruptcy prediction models. The proposed framework is a three-stage process and could be summarized as follows:

Stage 1 – Returns-to-scale (RTS) Analysis: Perform RTS analysis to find out whether to solve a DEA model under constant returns-to-scale (CRS) conditions, variable returns-to-scale (VRS) conditions, increased returns-to-scale (IRS) conditions, or decreased returns-to-scale (DRS) conditions – see Banker et al. (2004) for details.

Stage 2 – Classification of DMUs: For each DMU k (k = 1, …, n), solve the following slacks-based measure (SBM) model (Tone, 2001):

$$\begin{aligned} \text{Min} \quad & \rho_k = \left(1 - \frac{1}{m}\sum_{i=1}^{m}\frac{s_{i,k}^{-}}{x_{i,k}}\right) \Big/ \left(1 + \frac{1}{s}\sum_{r=1}^{s}\frac{s_{r,k}^{+}}{y_{r,k}}\right) \\ \text{s.t.:} \quad & \sum_{j=1}^{n}\lambda_j x_{i,j} + s_{i,k}^{-} = x_{i,k}; \forall i \\ & \sum_{j=1}^{n}\lambda_j y_{r,j} - s_{r,k}^{+} = y_{r,k}; \forall r \\ & \lambda_j \geq 0, \forall j; s_{i,k}^{-} \geq 0, \forall i; s_{r,k}^{+} \geq 0, \forall r \end{aligned} \tag{1}$$

where n denotes the number of DMUs, m is the number of inputs, s is the number of outputs, $x_{i,j}$ is the amount of input i used by DMU_j, $y_{r,j}$ is the amount of output r produced by DMU_j, λ_j is the weight assigned to DMU_j in constructing its ideal benchmark, and $s_{i,k}^{-}$ and $s_{r,k}^{+}$ are slack variables associated with the first and the second sets of constraints, respectively. If the optimal objective function value $\rho_k^* = 1$, then DMU_k is classified as efficient. If $\rho_k^* \prec 1$, DMU_k is classified as inefficient. Note that model 1 above is solved as it is if stage 1 reveals that the CRS conditions hold; otherwise, one would have to impose one of the following additional constraints depending on whether VRS, IRS, or DRS conditions prevail, respectively:

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$$\sum_{j=1}^{n}\lambda_j = 1;\ \sum_{j=1}^{n}\lambda_j \geq 1\ ;\ \sum_{j=1}^{n}\lambda_j \leq 1. \tag{2}$$

Stage 3 – Break Efficiency Ties: For each efficient DMU *k*, solve the following slacks-based super-efficiency DEA model – first proposed by Tone (2002):

$$\begin{aligned}
&\text{Min} \quad \delta_k = \left(\frac{1}{m}\sum_{i=1}^{m}\frac{\left(x_{i,k}+t_{i,k}^{-}\right)}{x_{i,k}}\right)\Bigg/\left(\frac{1}{s}\sum_{r=1}^{s}\frac{\left(y_{r,k}+t_{r,k}^{+}\right)}{y_{r,k}}\right)\\
&\text{s.t.:} \quad \sum_{j=1;\,j\neq k}^{n}\lambda_j x_{i,j} \leq x_{i,k}+t_{i,k}^{-};\forall i\\
&\qquad \sum_{j=1;\,j\neq k}^{n}\lambda_j y_{r,j} \geq y_{r,k}-t_{r,k}^{+};\forall r\\
&\qquad \lambda_j \geq 0,\forall j\neq k;\ t_{i,k}^{-}\geq 0,\forall i;\ t_{r,k}^{+}\geq 0,\forall r
\end{aligned} \tag{3}$$

where $t_{i,k}^{-}$ (respectively, $t_{r,k}^{+}$) denotes the amount by which input *i* (respectively, output *r*) of the efficient DMU_k should be increased (respectively, decreased) to reach the frontier constructed by the remaining DMUs. Note that the model 2 above is solved as it is if stage 1 reveals that the CRS conditions hold; otherwise, one would have to impose an additional constraint from amongst (2) as outlined in stage 2. Use the super-efficiency scores δ_k^*s to rank order the efficient DMUs.

At this stage, it is worth mentioning that unlike radial super-efficiency DEA models (e.g., Andersen and Petersen, 1993), slacks-based super-efficiency models are always feasible (Tone, 2002, Du et al., 2010). Note that Tone (2002) and Du et al. (2010) slacks-based super-efficiency models are identical with respect to their constraints in that one could be obtained from the other using a simple variable transformation. Note, however, that in applications where positive input and output data is a requirement, Du et al. (2010) provide a variant of the model solved in stage 3 to accommodate this situation. In the next section, we shall use the above described methodology to rank order competing bankruptcy prediction models and discuss the empirical results obtained using UK data for the period of 1989-2006.

5. Empirical Investigation

In this section, we shall first describe the process of data gathering and sample selection on the relative performance evaluation of bankruptcy prediction models under multiple criteria. Next, we present and discuss our empirical findings.

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5.1. Data

In this paper, we first considered all UK firms listed on the London Stock Exchange (LSE) during an 18 years period from 1989 through 2006 and defined the bankrupt ones using the London Share Price Database (LSPD) codes *16* (i.e., firm has receiver appointed or is in liquidation), *20* (i.e., firm is in administration or administrative receivership), and *21* (i.e., firm is cancelled and assumed valueless); the remaining firms are classified as non-bankrupt. Then, we further reduced such dataset by excluding both financial and utilities firms, on one hand, and those firms with less than 5 month lag between the reporting date and the fiscal year, on the other hand. As a result of these data reduction rules, the final sample consists of 1414 UK listed firms – 211 of which are bankrupt firms and the remaining 1203 are non-bankrupt firms. In sum, our sample consists of a total of 12452 firm-year observations including 2062 observations related to bankrupt firms and 10390 observations related to non-bankrupt firms. Within our sample, the average bankruptcy rate is 1.63% per year – which is higher than the 0.67% (respectively, 1.19%) bankruptcy rate of the sample used by Agarwal and Taffler (2008) (respectively,Christidis and Gregory, 2010). This higher rate of bankruptcy in comparison with other studies is due to the period of study being extended. Note however that the actual numbers of observations used to estimate the various models differ depending on the availability of data on each explanatory variable.

5.2. Model estimations

In this paper, we have chosen to assess the relative performance of the most widely used statistical models that have been implemented in the literature on bankruptcy prediction (see for example, Hillegeist et al., 2004, Agarwal and Taffler, 2008); namely, *MDA models* proposed by Altman (1968), Altman (1983), Taffler (1984) and Lis (1972), and *regression models* such as re-estimate four above mentioned MDA models in the logit framework, Ohlson (1980), Zmijewski (1984) and Bemmann (2005), and hazard model proposed by Shumway (2001), and *BSM-based models* proposed by Hillegeist et al. (2004) and Bharath and Shumway (2008) – see Appendix A for details. Note that all chosen models are tested out-of-sample and the training period ranges from 1989 to 2001 including 1571 failure and 5615 non-failure firm-year observations.

Table 3 reports the estimated discriminant weights and coefficients of reworked alternative bankruptcy models using training sample from 1989 through 2001. For MDA models the weights

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of the discriminant function which are corresponding to the difference of the mean vectors of the explanatory variables for the healthy and failed firms are reported. These weights are estimated such that the value of the discriminant function for two groups of healthy and failure firms differ as much as possible.

For probability models (e.g, logit and probit), the estimated coefficients of explanatory variables together with their adjusted Wald Chi-square are reported. Note that in the multi-period logit model, the number of independent observations is increased to the number of firm year observations which results in highly serial correlation between observation and an upward estimated standard error. As a result, according to Shumway (2001) the estimated standard errors should be adjusted, where the produced Wald Chi-square from the logit model is divided by the average number of firm-years. Finally, the coefficient of estimated discrete hazard model of Shumway (2001) in adjacent to their *z-statistic* are provided.

5.3. Performance evaluation of bankruptcy prediction models

In our empirical investigation, we first generated rankings for many measures of the same criterion for the sake of finding out about the extent to which rankings are robust to the choice of measures. Table 4 provides the unidimensional rankings of fourteen prediction models of predicting the UK bankrupted firms based on measures of four criteria; namely, discriminatory power for population behaviour, discriminatory power for average behaviour, calibration accuracy and correctness of categorical prediction, which is a typical output from existing studies (e.g., Hillegeist et al., 2004; Agarwal and Taffler, 2008; Bharath and Shumway 2008). These unidimensional rankings are devised as follows: models are ranked in the ascending (respectively, descending) order of the relevant measure of each of the criteria under consideration if the measure is to be minimized (respectively, maximized). One could clearly see that rankings could be different from one measure to another, thus resulting in a situation where one cannot make an informed decision as which model perform best with respect to all criteria. For example, Shumway (2001)'s Hazard model outperforms all four MDA models on measures of discriminatory power, correctness of categorical prediction, whereas Altman (1968), Altman (1983) and Lis (1972)'s MDA models performs better with respect to the calibration criterion, as measured by BS and HL. In order to address this issue, one would need a single ranking that takes account of multiple criteria, which we provide using the proposed DEA framework.

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Table 5 summarizes the multi-criteria rankings of fourteen competing bankruptcy prediction models using different combinations of measures for each criterion. The models are ranked in descending order of the corresponding SBM super efficiency DEA scores. The empirical result reveals that the multidimensional rankings have no ties, and multidimensional rankings differ from unidimensional ones. For example, the unidimensional rankings corresponding to AUC, CIER, HL and T1 are different, on one hand, and have ties, on the other hand, as compared to the multidimensional ranking corresponding to AUC, CIER, HL and T1, where ties have been resolved – see Table 5. Therefore, the proposed multidimensional framework provides a valuable tool to apprehend the true nature of the relative performance of bankruptcy prediction models. Furthermore, we have considered several measures of the population behaviour of discriminatory power criterion, calibration accuracy criterion, and the correctness of categorical predictions criterion to find out the robustness of multidimensional rankings with respect to different measures. We find that whether one measures population behaviour of discriminatory power by AUC, KS or IV, measure calibration accuracy by BS or HL, and measure correctness of categorical prediction by T1 or MR, the ranks of the best models such as Shumway (2001)'s hazard model, Hilligieski et al. (2004)'s BSM model remain the same. Thus, they are robust to changes in measures.

From a bankruptcy prediction perspective, the following observations and conclusions could be made. *First*, dynamic models systematically outperform the static one regardless of the choice of measures. *Second*, models that make use of market-based information tend to outperform those that make use of accounting-based information only, which could be explained by the fact that market data is supposed to incorporate historical data (e.g., accounting information), current information on firms, and eventually future expectations. *Third*, within the static models sub-category, probability models (i.e., logit, probit) tend to outrank DA models; in addition, reworking DA models within a logit framework results in a another category of models with average performance. *Fourth*, with respect to the discriminatory power criterion, the rankings of models are robust to choices of measures such as KS and ROC. *Fifth*, with respect to the sub-category of discriminatory power measures, the ranking of models seem to be sensitive to whether one penalizes Type I errors or Type II errors. However, due to the higher costs related to Type I errors compared to the costs related to Type II errors, most practitioners would prefer penalizing Type I errors. *Sixth*, with respect to the calibration criterion, the rankings seem to be

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more sensitive to the choice of measures, which could be justified by the fact that BS penalizes inaccuracy in a "heavier" way as compared to HL. This suggests that, in practice, one would have to select the most appropriate measure for the decision making situation under consideration. Last, but not least, notice that using simultaneously Type I errors and Brier Score as "over penalizing" measures produces results that are more conservative for bankers/loan providers, which tend to privilege Logit and Probit models in the rankings.

6. Conclusion

Prediction of corporate failures is one of the major activities in auditing firms' risks and uncertainties. The design of reliable models to predict bankruptcy is crucial for many decision making processes. Given the large number of models have been designed to predict bankruptcy; another key task is to evaluate the relative performance of prediction models and find out which one(s) provides the best forecasts. Although most studies tend to use several performance criteria, and for each criterion, one or several metrics to measure each criterion, the assessment exercise of the relative performance of prediction models is restricted to their ranking by measure, which often leads to conflicting results. In this research, we overcome this methodological issue by proposing an orientation-free data envelopment analysis (DEA) model as a multi-criteria assessment framework, which delivers a single ranking based on multiple performance criteria. We evaluate the relative performance evaluation of thirteen prediction models using UK firms listed on the LSE for the period from 1989 to 2006. We use three criteria which are commonly used in the literature; namely, the discriminatory power, calibration and correctness of categorical prediction criteria. We have considered several measures for each criterion to find out the robustness of multidimensional rankings with respect to different measures. The main conclusions of this research could be summarized as follows. First, the proposed multidimensional framework provides a valuable tool to apprehend the true nature of the relative performance of bankruptcy models. Second, the multidimensional rankings of the best and the worst models seem to be robust to changes in most performance. In particular, dynamic models systematically outperform the static one regardless of the choice of measures. Third, models that make use of market-based information tend to outperform those that make use of accounting-based information only.

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References

AGARWAL, V. & TAFFLER, R. 2008. Comparing the performance of market-based and accounting-based bankruptcy prediction models. *Journal of Banking & Finance,* 32**,** 1541-1551.

AHN, B. S., CHO, S. S. & KIM, C. Y. 2000. The integrated methodology of rough set theory and artificial neural network for business failure prediction. *Expert Systems with Applications,* 18**,** 65-74.

ALTMAN, E. 1983. Corporate Financial Distress: A Complete Guide to predicting, Avoiding and Dealing with Bankruptcy. *Edit. John Wiley and Sons*.

ALTMAN, E. I. 1968. Financial ratios, discriminant analysis and the prediction of corporate bankruptcy. *The journal of finance,* 23**,** 589-609.

ANDERSEN, P. & PETERSEN, N. C. 1993. A procedure for ranking efficient units in data envelopment analysis. *Management science,* 39**,** 1261-1264.

ARMINGER, G., ENACHE, D. & BONNE, T. 1997. Analyzing credit risk data: A comparison of logistic discrimination, classification tree analysis, and feedforward networks. *Computational Statistics,* 12.

BALCAEN, S. & OOGHE, H. 2006. 35 years of studies on business failure: an overview of the classic statistical methodologies and their related problems. *The British Accounting Review,* 38**,** 63-93.

BANDYOPADHYAY, A. 2006. Predicting probability of default of Indian corporate bonds: logistic and Z-score model approaches. *The Journal of Risk Finance,* 7**,** 255-272.

BEAVER, W. H. 1966. Financial ratios as predictors of failure. *Journal of accounting research***,** 71-111.

BEMMANN, M. 2005. Improving the comparability of insolvency predictions.

BHARATH, S. T. & SHUMWAY, T. 2008. Forecasting default with the Merton distance to default model. *Review of Financial Studies,* 21**,** 1339-1369.

BOSER, B. E., GUYON, I. M. & VAPNIK, V. N. A training algorithm for optimal margin classifiers. 1992 1992. ACM, 144-152.

BREIMAN, L., FRIEDMAN, J. H. & OLSHEN, R. A. 1984. Classification and Regression Trees. *Wadsworth, Belmont, California*.

BRYANT, S. M. 1997. A case-based reasoning approach to bankruptcy prediction modeling. *Intelligent Systems in Accounting, Finance and Management,* 6**,** 195-214.

CHARNES, A. & COOPER, W. W. 1962. Programming with linear fractional functionals. *Naval Research logistics quarterly,* 9**,** 181-186.

CHARNES, A., COOPER, W. W. & RHODES, E. 1978. Measuring the efficiency of decision making units. *European Journal of Operational Research,* 2**,** 429-444.

CHRISTIDIS, A. & GREGORY, A. 2010. Some new models for financial distress prediction in the UK. *Xfi-Centre for Finance and Investment Discussion Paper*.

COOK, W. D. & SEIFORD, L. M. 2009. Data envelopment analysis (DEA)–Thirty years on. *European Journal of Operational Research,* 192**,** 1-17.

COOPER, W. W., SEIFORD, L. M., THANASSOULIS, E. & ZANAKIS, S. H. 2004. DEA and its uses in different countries. *European Journal of Operational Research,* 154**,** 337-344.

COX, D. R. 1972. Regression models and life-tables. *Journal of the Royal Statistical Society. Series B (Methodological)***,** 187-220.

Electronic copy available at: https://ssrn.com/abstract=3802750

DIMITRAS, A. I., SLOWINSKI, R., SUSMAGA, R. & ZOPOUNIDIS, C. 1999. Business failure prediction using rough sets. *European Journal of Operational Research,* 114**,** 263-280.

DU, J., LIANG, L. & ZHU, J. 2010. A slacks-based measure of super-efficiency in data envelopment analysis: a comment. *European Journal of Operational Research,* 204**,** 694-697.

FISHER, R. A. 1938. The statistical utilization of multiple measurements. *Annals of Human Genetics,* 8**,** 376-386.

FREED, N. & GLOVER, F. 1981. A linear programming approach to the discriminant problem. *Decision Sciences,* 12**,** 68-74.

HEKANAHO, J., BACK, B., SERE, K. & LAITINEN, T. 1998. Analysing bankruptcy data with multiple methods. *Turku: American Association for Artificial Intelligence*.

HILLEGEIST, S. A., KEATING, E. K., CRAM, D. P. & LUNDSTEDT, K. G. 2004. Assessing the probability of bankruptcy. *Review of Accounting Studies,* 9**,** 5-34.

LIU, J. S., LU, L. Y. Y., LU, W.-M. & LIN, B. J. Y. 2013. Data envelopment analysis 1978–2010: A citation-based literature survey. *Omega,* 41**,** 3-15.

MARTIN, D. 1977. Early warning of bank failure: A logit regression approach. *Journal of Banking & Finance,* 1**,** 249-276.

MCDONALD, R. L., CASSANO, M. & FAHLENBRACH, R. 2006. *Derivatives markets,* Addison-Wesley Boston.

OHLSON, J. A. 1980. Financial ratios and the probabilistic prediction of bankruptcy. *Journal of accounting research***,** 109-131.

PREMACHANDRA, I. M., BHABRA, G. S. & SUEYOSHI, T. 2009. DEA as a tool for bankruptcy assessment: A comparative study with logistic regression technique. *European Journal of Operational Research,* 193**,** 412-424.

RAVI KUMAR, P. & RAVI, V. 2007. Bankruptcy prediction in banks and firms via statistical and intelligent techniques–A review. *European Journal of Operational Research,* 180**,** 1-28.

SHUMWAY, T. 2001. Forecasting Bankruptcy More Accurately: A Simple Hazard Model*. *The Journal of Business,* 74**,** 101-124.

TAFFLER, R. J. 1984. Empirical models for the monitoring of UK corporations. *Journal of Banking & Finance,* 8**,** 199-227.

TAM, K. Y. & KIANG, M. Y. 1992. Managerial applications of neural networks: the case of bank failure predictions. *Management science,* 38**,** 926-947.

TONE, K. 2002. A strange case of the cost and allocative efficiencies in DEA. *Journal of the Operational Research Society,* 53**,** 1225-1231.

VAN GESTEL, T., BAESENS, B., VAN DIJCKE, P., GARCIA, J., SUYKENS, J. A. K. & VANTHIENEN, J. 2006. A process model to develop an internal rating system: sovereign credit ratings. *Decision Support Systems,* 42**,** 1131-1151.

ZHU, J. 2003. *Quantitative models for performance evaluation and benchmarking: data envelopment analysis with spreadsheets and DEA excel solver,* Springer.

ZMIJEWSKI, M. E. 1984. Methodological issues related to the estimation of financial distress prediction models. *Journal of Accounting Research***,** 59-82.

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Table 1: Basic DEA Multiplier Models

Input-Oriented	**Output-Oriented**
$Maximize \quad e_k^{input} = \sum_{r=1}^{s} u_r y_{r,k}$	$Minimize \quad e_k^{output} = \sum_{i=1}^{m} v_i x_{i,k}$
$s.t.: \quad \sum_{i=1}^{m} v_i x_{i,k} = 1$	$s.t.: \quad \sum_{r=1}^{s} u_r y_{r,k} = 1$
$\sum_{r=1}^{s} u_r y_{r,j} \le \sum_{i=1}^{m} v_i x_{i,j}; \; j = 1,...,n$	$\sum_{r=1}^{s} u_r y_{r,j} \le \sum_{i=1}^{m} v_i x_{i,j}; \; j = 1,...,n$
$v_i \ge 0; i = 1,...,m$	$v_i \ge 0; i = 1,...,m$
$u_r \ge 0; r = 1,...,s$	$u_r \ge 0; r = 1,...,s$

Table 2: Basic DEA Envelopment Models

Input-Oriented	**Output-Oriented**
$Minimize \quad \theta_k$	$Maximize \quad \phi_k$
$s.t.: \quad \sum_{j=1}^{n} \lambda_j x_{i,j} \le \theta_k \cdot x_{i,k}; \forall i$	$s.t.: \quad \sum_{j=1}^{n} \lambda_j x_{i,j} \le x_{i,k}; \forall i$
$\sum_{j=1}^{n} \lambda_j y_{r,j} \ge y_{r,k}; \forall r$	$\sum_{j=1}^{n} \lambda_j y_{r,j} \ge \phi_k \cdot y_{r,k}; \forall r$
$\lambda_j \ge 0; \forall j$	$\lambda_j \ge 0; \forall j$
θ_k *unrestricted*	ϕ_k *unrestricted*

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Table 3: Model Estimation results.The table presents the estimated parameters of the alternative bankruptcy prediction models from various econometric frameworks (e.g., MDA, Logit, Probit and Hazard models). The training sample is from 1989 to 2001 including 1571 failure and 5615 non-failure firm-year observations. The adjusted Wald Chi-Square (z) statistics from the logistic regression (proportional hazard regression) are presented together with each estimated parameter. A star, dragger and double dragger denote statistical significance at 1%, 5% and 10% level, respectively.

	Panel A: MDA Models				Panel B: Logit/Probit Models												Panel C: Hazard Model	
	Altman (1968)	Altman (1983)	Taffler (1984)	Lis (1972)	Ohlson (1980)		Zmigeski (1984)		Altman (1968)		Altman (1983)		Taffler (1984)		Lis (1972)		Bemmann (2005)		Shumway (2001)	
	Estimate	Estimate	Estimate	Estimate	Estimate	Wald ChiSqu.	Estimate	Wald ChiSqu.	Estimate	Wald ChiSqu.	Estimate	Wald ChiSqu.	Estimate	Wald ChiSqu.	Estimate	Wald ChiSqu.	Estimate	Wald ChiSqu	Estimate	Z
Intercept	0.485	0.396	1.646	-0.148	**-3.842**	16.08‡	**-2.221**	134.96*	**-3.831**	129.58*	**-3.845**	95.84*	**-4.518**	106.70*	**-3.701**	191.68*	**-4.471**	149.71*		
Profitability variables																				
EBITTA	3.420	2.503		1.940					**-1.412**	9.16*	**-0.939**	4.89*			-0.592	2.81‡				
SALETA	-0.278	-0.233							0.171	0.77	0.104	0.24								
NITA					0.535	0.814	**-0.291**	3.82*											**0.523**	9.432*
CHIN					**-1.149**	8.33*														
PBTCL			0.480										**-0.292**	8.83*						
Liquidity variables																				
WCTA	3.633	3.041		2.694	**-3.615**	8.33*	-0.027	0.215	**-2.799**	14.51*	**-2.016**	8.81*			**-2.114**	6.84*				
CLCA*					-0.373	1.830							-0.144	0.62						
FUTL					-0.401	3.59*														
INTWO					**1.282**	5.71†														
CLTA			-3.176										**1.611**	9.52*						
NCI			0.000										0.000	0.022						
NWTA				0.793											0.1633	0.09				
Leverage variables																				
RETA	-0.387	-0.242							0.387	3.61‡	0.203	1.60								
METL	0.000			0.000					0.000	0.23					0.000	0.020				
BVETL		-0.009									-0.025	0.06								
TLTA					0.895	1.230	0.313	2.480									**0.986**	6.46*	**1.299**	66.67*
OENEG					-1.431	1.750														
TLMTA																				
CATL			0.095																	
Size Variable																				
Ohlsonsize					-0.144	0.350														
Relative Size																			**-0.312**	39.05*
Other firm-characteristics																				
LagExReturn																			**-1.337**	105.24*
LagSigma																			0.431	0.2205

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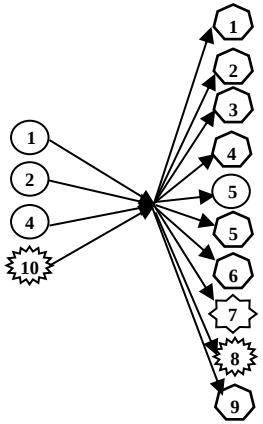
Table 4: Unidimensional Rankings of Bankruptcy Prediction Models - Ranking in Descending Order of A Specific Performance Measure.
The table presents the best to worst ranking of fourteen competing bankruptcy models using a unidimensional ranking framework. The alternative models are ranked based on variety of measures related to different criteria. T1 (type I error), T2 (type II error), MR (misclassification rate) and OCC (overall correct classification) are used as measures for correctness of categorical prediction; AUC (area under receiver operating character), gini coefficient , KS (Kulmogrov Smirnove) and IV (information value) are used as measures of separation between functions of failure and non-failure firms; CIER (conditional information entropy ratio) is implemented as measure of discrimination between the expected distribution of failure and non-failure firms; BS (brier score) and HL(Hosmer Lemeshow) are considered as measures of calibration. Different shapes represents different econometric framework for reworking models. Note that some models are reworked in two different frameworks.

Criterion	Measures	Rank from Best to Worst
Correctness of categorical prediction	T1	7 → 3 → 2 → 6 → 5 → 2 → 1 → 8 → 4 → 5 → 4 → 10 → 1 → 9
	T2, MR & OCC	7 → 8 → 10 → 5 → 5 → 4 → 9 → 1 → 4 → 2 → 1 → 3 → 6 → 10
Discriminatory Power: Separation	AUC, Gini	7 → 2 → 8 → 3 → 6 → 5 → 1 → 4 → 5 → 2 → 10 → 4 → 9 → 1
	KS	7 → 8 → 2 → 3 → 1 → 6 → 5 → 4 → 1 → 10 → 2 → 5 → 4 → 9
	IV	8 → 10 → 7 → 3 → 1 → 2 → 9 → 4 → 5 → 1 → 6 → 2 → 4 → 5
Discriminatory Power: Expectation	CIER	8 → 10 → 4 → 5 → 9 → 1 → 3 → 2 → 7 → 6 → 1 → 4 → 5 → 2
Calibration	BS	9 → 6 → 5 → 3 → 4 → 5 → 4 → 7 → 8 → 2 → 2 → 10 → 1 → 1
	HL	

[1]Altman (1968) ; [2]Lis (1972); [3] Ohlson (1980); [4]Altman (1983); [5] Taffler (1984); [6] Zmijewski (1984); [7] Shumway (2001); [8] Hilligieski et.al (2004); [9]Bemmann (2005); [10]Baharath & Shumway (2008)

◯ = MDA model; ⬡ = Probability model; ⬠ = Survival analysis model; ✸ = BSM-based model

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Table 5: Super Efficiency DEA model scores-based multidimensional rankings of volatility forecasting models

Inputs	Outputs	Rank from Best to Worst
BS; T1	AUC; CIER	8 → 7 → 9 → 3 → 6 → 5 → 4 → 10 → 2 → 1 → 5 → 4 → 2 → 1
BS; T1	KS; CIER	8 → 7 → 3 → 9 → 6 → 5 → 4 → 10 → 2 → 1 → 5 → 4 → 2 → 1
BS; T1	IV; CIER	8 → 7 → 3 → 9 → 6 → 5 → 4 → 10 → 2 → 1 → 5 → 4 → 2 → 1
BS; MR	AUC; CIER	8 → 7 → 9 → 6 → 3 → 5 → 4 → 10 → 1 → 2 → 5 → 4 → 1 → 2
BS; MR	KS; CIER	8 → 7 → 9 → 6 → 3 → 5 → 4 → 10 → 1 → 2 → 5 → 4 → 1 → 2
BS; MR	IV ; CIER	8 → 7 → 9 → 3 → 5 → 4 → 10 → 1 → 2 → 6 → 5 → 4 → 1 → 2
HL; T1	AUC; CIER	8 → 7 → 10 → 2 → 3 → 5 → 4 → 2 → 9 → 1 → 6 → 1 → 4 → 5
HL; T1	KS; CIER	8 → 7 → 10 → 2 → 3 → 5 → 4 → 2 → 9 → 1 → 6 → 1 → 4 → 5
HL; T1	IV; CIER	8 → 7 → 10 → 2 → 3 → 5 → 4 → 2 → 9 → 1 → 6 → 1 → 4 → 5
HL; MR	AUC; CIER	8 → 7 → 10 → 2 → 4 → 5 → 9 → 1 → 3 → 2 → 6 → 1 → 4 → 5
HL; MR	KS; CIER	8 → 7 → 10 → 2 → 4 → 5 → 9 → 1 → 3 → 2 → 1 → 6 → 4 → 5
HL; MR	IV ; CIER	8 → 7 → 10 → 4 → 5 → 9 → 1 → 3 → 2 → 6 → 1 → 4 → 2 → 5

[1]Altman (1968) ; [2]Lis (1972); [3] Ohlson (1980); [4]Altman (1983); [5] Taffler (1984); [6] Zmijewski (1984); [7] Shumway (2001); [8] Hilligieski et.al (2004); [9]Bemmann (2005); [10]Baharath & Shumway (2008)

◯ = MDA model; ⬡ = Probability model; ⬠ = Survival analysis model; ✹ = BSM-based model

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Appendix A: Statistical Models of Bankruptcy Prediction

Model/ Type	Variables		Formula / Description
Altman (1968) *Discriminant Analysis*	WCTA RETA EBITTA METD STA	= working capital/total assets, = retained earnings/total assets, = Earnings before interest and taxes/total assets, = Market value of equity/total liabilities, = Sales/total assets.	The formula for estimating all MDA models is: $$z = f\left(\sum_{j=1}^{p} \beta_j x_j\right)$$ where z is commonly referred to as a score, x_j s are explanatory variables, β_j s represent the coefficients of the explanatory variables in the model, and f denotes the mapping of $\beta^t \cdot x$ on the set of real numbers $\Re$. Theses multivariate linear discriminant analysis (MDA) models were proposed by Altman (1968, 1983) to classify firms into two groups; namely, failing firms and non-failing firms, based on five ratios. The latter (Altman, 1983) is the revised version of the former model (Altman, 1968) by substituting the book values of equity for the market value. Both of models were driven from US market data. Both Lis (1972) and Taffler (1984) are MDA models in where UK market data is used for estimating.
Altman (1983) *Discriminant Analysis*	WCTA RETA EBITTA BETL STA	=Working capital/total assets, =Retained earnings/total assets, =Earnings before interest and taxes/total assets, =Book value Equity/total liabilities, =Sales/total assets.	
Lis (1972) *Discriminant Analysis*	WCTA EBITTA METL NWTA	=Working capital/total assets, =Earnings before interest and taxes/total assets, =Market value of equity/total liabilities, =Net working capital/total asset.	
Taffler (1984) *Discriminant Analysis*	PBTCL CLTA CATL NCI	=Profit before tax/current liabilities, =Current liabilities/total assets, =Current assets/total liabilities, =No-credit intervals.	
Ohlson (1980) *Logit Model*	TLTA WCTA CLCA OENEG NITA FUTL INTWO CHIN Ohlsonsize	=Total liabilities/total assets =Working capital/total assets =Current liabilities/current assets =0 if total liabilities exceed total assets, 1 otherwise =Net income/total assets = Funds from operations (operating income minus depreciation)/total liabilities =1 if net income has been negative for the last 2 years, 0 otherwise, = $NI_t - NI_{t-1} / (\lvert NI_t \rvert + \lvert NI_{t-1} \rvert)$, NI_t is net income for the last period. The variable is a proxy for the relative change in net income. = log(total assets/GNP price-level index)	The formula for the logistic regression is: $$PD = \Lambda\left(\beta^t x\right) = \frac{e^{\beta^t x}}{1 + e^{\beta^t x}}$$ where PD is the probability of default bounded between 0 and 1, and Λ is the function to map the score ($\beta^t \cdot x$) to the logistic probability. It is the pioneer model which uses a cumulative logistic distribution as a link to change the scores to a probability.
Zmijewski (1984) *Probit Model*	NITA TLTA CACL	=Net income /total assets =Total liabilities/total assets =Current assets/current liabilities	A pioneer model in which a cumulative normal distribution is used as a link function to map the score ($\beta^t \cdot x$) to the probit probability between 0 and 1. The formula for probit model is: $PD = \Phi\left(\beta^t x\right) = \int_{-\infty}^{\beta^t x} \varphi(t)\, dt$ where Φ and φ refer to cumulative and normal distribution respectively.

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Model/ Type	Variables		Formula / Description
Bemmann (2005) *Logit model*	TLTA	=Total Liabilities /total assets	A univariate explanatory variable model which is estimated in a logit framework.
Shumway (2001) *Hazard Model*	NITL TLTA RelSize LagExRet LagSigma	=Net income/total liabilities, =Total liabilities/total assets, =Log (the number of outstanding shares multiplied by year-end share price divided by total market value), =Cumulative annual return in year t-1 minus the value-weighted FTSE index return in year t -1, =Standard deviation of the residual derived from regressing monthly stock return on market return in year t -1.	The hazard model is estimated as follows: $PD_{i,t} = \Lambda(y_{i,t})$ $y_{i,t} = \alpha + \beta' X_{i,t-1} = \beta' \begin{pmatrix} X_{1,t-1} & \cdots & X_{1,t-j} \\ \vdots & \ddots & \vdots \\ X_{n,t-1} & \cdots & X_{n,t-j} \end{pmatrix}$ where *n* represents the number of explanatory variables and *j* refers to the number of time periods before time *t*. Due to the same likelihood function (Shumway, 2001) a hazard model for probability of default is replaced with a logit model in which an information matrix including all historical values of the explanatory variables for a firm is used instead of each firm-year as an independent observation.
Hillegeist et al. (2004) & Bharath and Shumway (2008) *Black-Scholes Option-pricing model*	V_E V_A μ δ X σ_A T	=Market value of equity, =Market value of assets, =Continuously compounded expected return on assets, =Continuous dividend rate expressed in terms of V_A, =Face value of debt maturing at time T, =Asset volatility, =Time to debt maturity, considers as 1 year.	Both BSM based models of HKCL and BS use market based information to estimate the probability of default: $PD = N\left(-\frac{\ln(V_A/X) + (\mu - \delta - o.5\sigma_A^2)T}{\sigma_A\sqrt{T}}\right)$ For HKCL, V_A and σ_A are estimated by solving these two equations simultaneously: $V_E = V_A e^{-\delta T} N(d_1) - X e^{-rT} N(d_2) + (1 - e^{-\delta T}) V_A,$ $\sigma_E = \frac{V_A e^{-\delta T} N(d_1)\sigma_A}{V_E},$ The starting value for V_A and σ_A is $V_A + X$ and $\sigma_E V_E/(V_E + X)$ respectively, and μ is estimated by; $\mu = \left(\frac{V_{A,t} - V_{A,t-1} + dividends}{V_{A,t-1}}\right)$ For BS, V_A and σ_A are estimated by following equations; $V_A = V_E + X,$ $\sigma_A = \frac{V_E}{V_A}\sigma_E + \frac{X}{V_A}\sigma_D,$ $\sigma_D = 0.05 + 0.25 \times \sigma_E.$ and μ is peroxided by the previous year stock return. The value of μ is bounded between risk free rate and 100%.

[1]HKCL stands for the model proposed by Hillegiest et.al. (2004); [2] BS stands for the model proposed by Bharah and Shumway (2008)

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Appendix B: Performance Measures for Assessing Bankruptcy Prediction Models

Criteria	Measure	Formula	Definition
Discriminatory Power	KL distance	$KL = H(p) - H_S$ where $H(p) = -\big(p \cdot log(p) + (1-p) \cdot log(1-p)\big)$ $H\big(P(D\|S)\big) = -(P(D\|S) \log P(D\|S) + P(ND\|S) \log P(ND\|S))$	**Kullback-Leibler (KL) distance** – also referred to as the divergence statistic, is a measure of entropy; that is, the expected value of the information content associated with the outcome of a model. It is the difference between the unconditional entropy $H(p)$ and the conditional entropy H_S. Note that **the** higher the KL statistic value, the better is the model under evaluation at reducing information uncertainty.
	CIER	$CIER_S = \frac{H(P) - H_S}{H(P)}$	The **conditional information entropy ratio (CIER)** is similar to KL distance except that it expressed KL in percentage terms. Notice that the closer the CIER value to one, the more information about the default event is contained in the rating score S.
	IV	$IV = \sum_{i=1}^{I}(g_i/n_G - b_i/n_B) \ln\left(\frac{g_i/n_G}{b_i/n_B}\right)$ where g_i (respectively, b_i) denotes the number of goods (respectively, bads) in band I and $\sum_{i=1}^{I} g_i = n_G$ and $\sum_{i=1}^{I} b_i = n_B$.	The **information value** (IV) is nothing but the **divergence statistic** originally suggested by Kullback and Leibler (1951) as a measure of the relative distance between an empirical probability distribution and a theoretical one. The measure provided here is based on a discrete approximation of the density functions of the good cases and the bad cases.
	D_M	$D = D_M^2$ where $D_M = \left\|\frac{\mu_G - \mu_B}{\sigma}\right\|$	The **Mahalanobis distance (D_M)** is an information value measure under the assumptions that the scores for the good cases and the bad cases are normally distributed and their variances are equal. In sum, D_M is the distance between the means of the conditional score distributions of the good cases and the bad cases, which when they are normal is also the mode (the highest value of the density function) divided by the standard deviation. Although this measure is popular, it remains that the validity of the information it conveys depends heavily on whether the underlying assumptions hold.
	Stability Index	$F = \sum_{i=1}^{n}\left[\left(\frac{O_i}{\sum O} - \frac{E_i}{\sum E}\right) \times Ln\left(\frac{O_i}{\sum O}/\frac{E_i}{\sum E}\right)\right]$ where O and E denote the observed and the expected frequencies, respectively, and n denotes the number of attributes.	In practice, one would be interested in comparing the bad cases distribution (respectively, good cases distribution) at two different points in time – in this case, the measure of the difference between these distributions is commonly referred to as the population stability index or simply the **stability index**. The result will always be positive, and a traffic-light approach is used to provide warnings: (i) green, less than 0.10, no cause for concern; (ii) yellow, between 0.10 and 0.25, some cause for concern, and (iii) red, greater than 0.25, concern. Note that the stability index is a special case of the divergence measure proposed by Kullback and Leibler (1951).
	KS statistic	$KS = \max_s\big(F(s\|B) - F(s\|G)\big)$, Where $F(.\|G)$ and $F(.\|B)$ denote the empirical cumulative distribution functions of the sample of good cases and the sample of bad cases, respectively.	The **Kolmogorov Smirnov (KS) statistic** measures the distance between the empirical cumulative distribution functions of the samples of good cases and bad cases.

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Discriminatory Power	AUC	$C - statistic = \frac{U}{N_G \cdot N_B}$ where U denotes the Mann-Whitney U statistic, N_G denotes the number of good cases in the sample, and N_B denotes the number of bad cases in the sample	The receiver operating characteristic (ROC) curve could also be seen as a plot of the hit rate against the false alarm rate for all cut-off points. A good prediction model would generate a ROC curve well farther away from the diagonal, which suggests that the larger the **area under the ROC curve**, often referred to as **AUC or AUROC** and measured by a statistic called concordance or **c-statistic**, the better the prediction model performance.
	Gini Coefficient	$G = 2 \cdot AUC - 1$	The **Gini coefficient**, say G, is a measure of the area between the curve and the diagonal – also referred to as **Somer's D**.
	Pietra Index	$Pietra\ Index = \frac{\sqrt{2}}{4} \max_c \lvert HR(C) - FAR(C) \rvert$	The **Pietra Index** is another statistic related to the ROC curve that measures half the maximal distance between the ROC curve and the diagonal.
	AR	$AR = \frac{Area\ between\ CAP\ curve\ and\ diagonal}{0.5(1 - P_B)}$ where P_B denotes the probability of bad	The **Cumulative Accuracy Profile (CAP)** curve plots the empirical cumulative distribution function (CDF) of the sample of bad cases against the empirical CDF of the whole sample including good and bad cases. The **Accuracy Ratio (AR)** is the ratio of the area between the CAP curve of the rating model being validated and the CAP curve of the random model, and the area between the CAP curve of the perfect rating model and the CAP curve of the random model. Note that, although the ROC and the CAP are not equivalent, the Gini index and the AR are equal for any scoring model.
Calibration Accuracy	Brier Score	$B = \frac{1}{N_G + N_B} \sum_{i=1}^{N_G+N_B} (Observed\ PD_i - Predicted\ PD_i)^2$	The Brier Score (B) is the mean squared error of probabilities of default
	Hosmer-Lemeshow statistic	$HL = \sum_{k=1}^{L} \frac{N_k \cdot (Observed\ PD_k - Predicted\ PD_k)^2}{Predicted\ PD_k \cdot (1 - Predicted\ PD_k)}$ where L denotes the number of credit scores' classes, N_k denotes the number of cases in class k.	The Hosmer-Lemeshow test is concerned with testing whether the predicted default probabilities and the observed default probabilities or realized default rates are identically distributed and the statistic used is referred to as the Hosmer-Lemeshow (HL) statistic. Notice that this test divides cases into a number of categories or classes depending on their scores.
	Binomial	$z = \frac{k - n\hat{p}}{\sqrt{n\hat{p}(1 - \hat{p})}}$ where n denotes the total number of cases in the sample, k denotes the observed number of bad cases and $\hat{p}$ is the predicted number of bad cases	In the context of bankruptcy prediction, the **Binomial test** is concerned with testing whether the predicted and the observed probabilities of default are identically distributed. Note however that the Normal approximation is typically used to avoid computational burden.
Correctness of Categorical	Sensitivity	$S_e = \frac{N_{B\|B}}{N_{B\|B} + N_{B\|G}}$ where $N_{B\|B}$ denotes the number of bad cases predicted as bad, $N_{B\|G}$ denotes the number of bad cases predicted as good	Given a specific cut-off score, say s_c, **sensitivity** (S_e) is defined as the fraction of the bad cases that would have scores below the cut-off and would therefore be rightly rejected; i.e., the proportion of bad cases who are predicted as bad. Sensitivity is also referred to as the **hit rate**.

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Specificity	$S_p = \frac{N_{G\|G}}{N_{G\|G}+N_{G\|B}}$ where $N_{G\|G}$ denotes the number of good cases predicted as good, $N_{G\|B}$ denotes the number of good cases predicted as bad	Given a specific cut-off score, say s_c, **specificity** (S_p) is defined as the fraction of the good cases that have scores above the cut-off and would therefore be rightly accepted; i.e., the proportion of good cases who are predicted as good.
Type I Error	Type I error = 1- Specificity	Type I error is the proportion of bad cases being misclassified as good cases
Type II Error	Type II error = 1- Sensitivity	Type II error is the proportion of good cases being misclassified as bad cases – also referred to as the false alarm rate.

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